Independent validation report: Retail Application Scorecard (RCS-APP-02)

Model under test: redeveloped/credit_scorecard/scorecard.py (Python rebuild, sha256 dc0f6c4e40ed, approved at Gate 3)
Generated: 2026-11-08T22:19:36.866884+00:00
Status: pending Gate 4
Re-validation triggered by monitoring: month 9 rated AMBER (2 consecutive amber months): Largest characteristic CSI 0.134 (AMBER, lim

OVERALL RATING: RED - Not fit for use until the red findings are remediated.

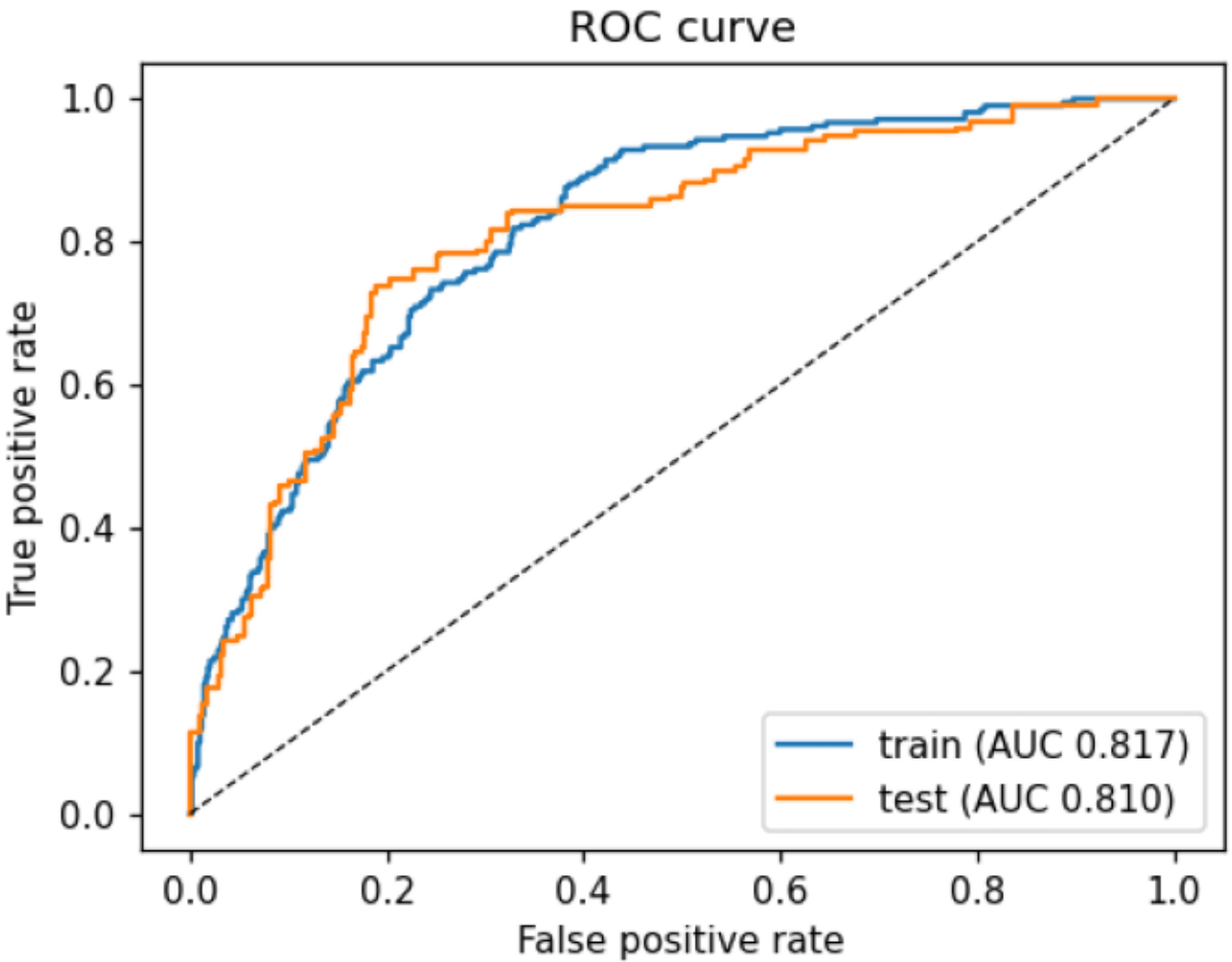
Ratings by area: performance GREEN, calibration RED, stability AMBER, fairness AMBER

Executive summary:
This re-validation, triggered by two consecutive amber monitoring months, rates the Retail Application Scorecard (RCS-APP-02) red overall, driven entirely by calibration. Discrimination remains strong, with a test AUC of 0.810 and a train-to-test drop of only 0.007, so the rank-ordering ability of the model is intact. However, the Hosmer-Lemeshow p-value of 2.65e-11 shows the predicted PDs no longer fit observed outcomes, with a mean predicted PD of 0.331 against an observed rate of 0.391. Stability and fairness are amber, reflecting characteristic-level drift and an age-group equalized odds gap. The model should not be relied on for PD-level decisions until it is recalibrated and the drift is investigated.

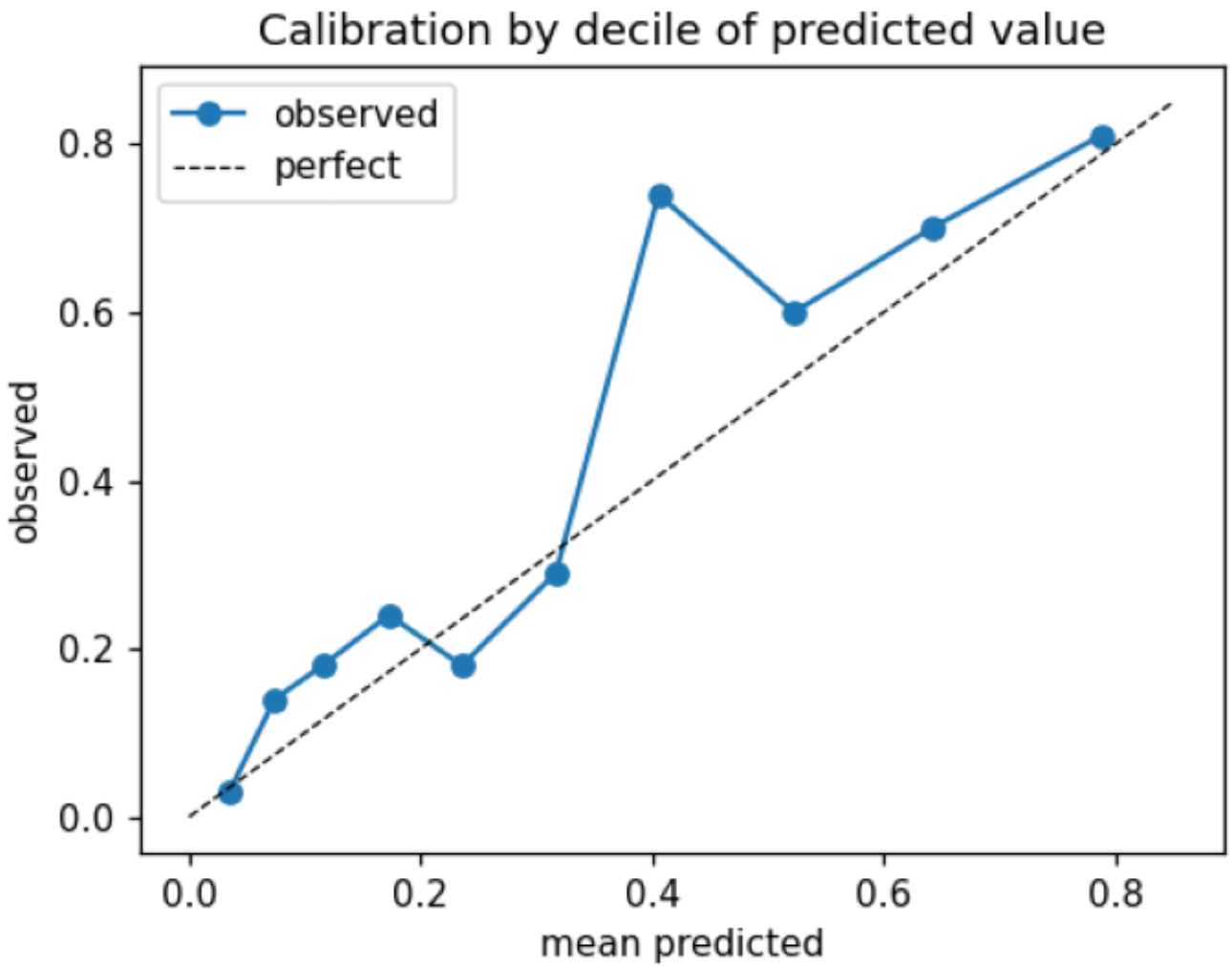
Checks:
[GREEN] performance: Discrimination: test AUC = 0.810 (>= 0.75 green, >= 0.65 amber) 95% bootstrap CI 0.782-0.835
[GREEN] performance: Overfitting: train AUC - test AUC = 0.007 (<= 0.03 green, <= 0.07 amber)
[AMBER] calibration: Calibration in the large: |mean predicted / observed - 1| = 0.154 (<= 0.1 green, <= 0.2 amber)
[RED] calibration: Hosmer-Lemeshow goodness of fit (p-value) = 2.65e-11 (>= 0.05 green, >= 0.01 amber) 10 equal-count groups, 8 d.f.
[GREEN] stability: Score PSI (train -> test) = 0.047 (<= 0.1 green, <= 0.25 amber)
[AMBER] stability: Characteristic stability: largest CSI = 0.134 (<= 0.1 green, <= 0.25 amber) variable duration
[AMBER] stability: Synthetic out-of-time: worst tranche score PSI = 0.189 (<= 0.1 green, <= 0.25 amber) 3 tranches of the test set in id order (synthetic time)
[GREEN] fairness: sex: adverse impact ratio of approval rates = 0.960 (>= 0.8 green, >= 0.7 amber)
[GREEN] fairness: sex: equalized odds difference = 0.095 (<= 0.1 green, <= 0.2 amber)
[GREEN] fairness: age_group: adverse impact ratio of approval rates = 0.908 (>= 0.8 green, >= 0.7 amber)
[AMBER] fairness: age_group: equalized odds difference = 0.154 (<= 0.1 green, <= 0.2 amber)
[AMBER] fairness: foreign_worker: group too small to rate = n/a (n >= 30 per group) A202 n=29

Findings:
[high] Predicted PDs are materially mis-calibrated on the monitoring batch: the Hosmer-Lemeshow p-value of 2.65e-11 is a red result, and mean predicted PD (0.331) is below the observed rate (0.391), with a deviation of 0.154 that is amber. -> Recalibrate the score-to-PD mapping (for example intercept or slope adjustment) on recent data, and restrict use of absolute PD outputs in pricing and provisioning until the recalibrated model is re-validated. Investigate whether the miscalibration reflects population drift or a model deficiency.
[medium] Characteristic drift is above the green limit (largest CSI 0.134) and the worst synthetic tranche score PSI is 0.189, both amber, though overall score PSI is 0.047. -> Identify the characteristic with the highest CSI, establish the cause of the drift (population change versus data or process issue), and assess its contribution to the calibration shortfall. Keep monthly CSI monitoring in place and consider redevelopment if drift persists after recalibration.
[medium] The age-group equalized odds difference of 0.154 is amber, indicating differing approval rates across age groups among applicants with the same true outcome. Age is a direct model input. -> Review the contribution of age-related characteristics to the score and document the business and statistical justification for their use. Assess the effect of removing or constraining them on both discrimination and the equalized odds gap, and continue monitoring.
[medium] The foreign worker attribute could not be tested reliably because one group has only 29 records, below the minimum of 30. -> Obtain a larger sample, or pool several months of data, so that the foreign worker group meets the minimum size and the test can be completed at the next validation or monitoring cycle.

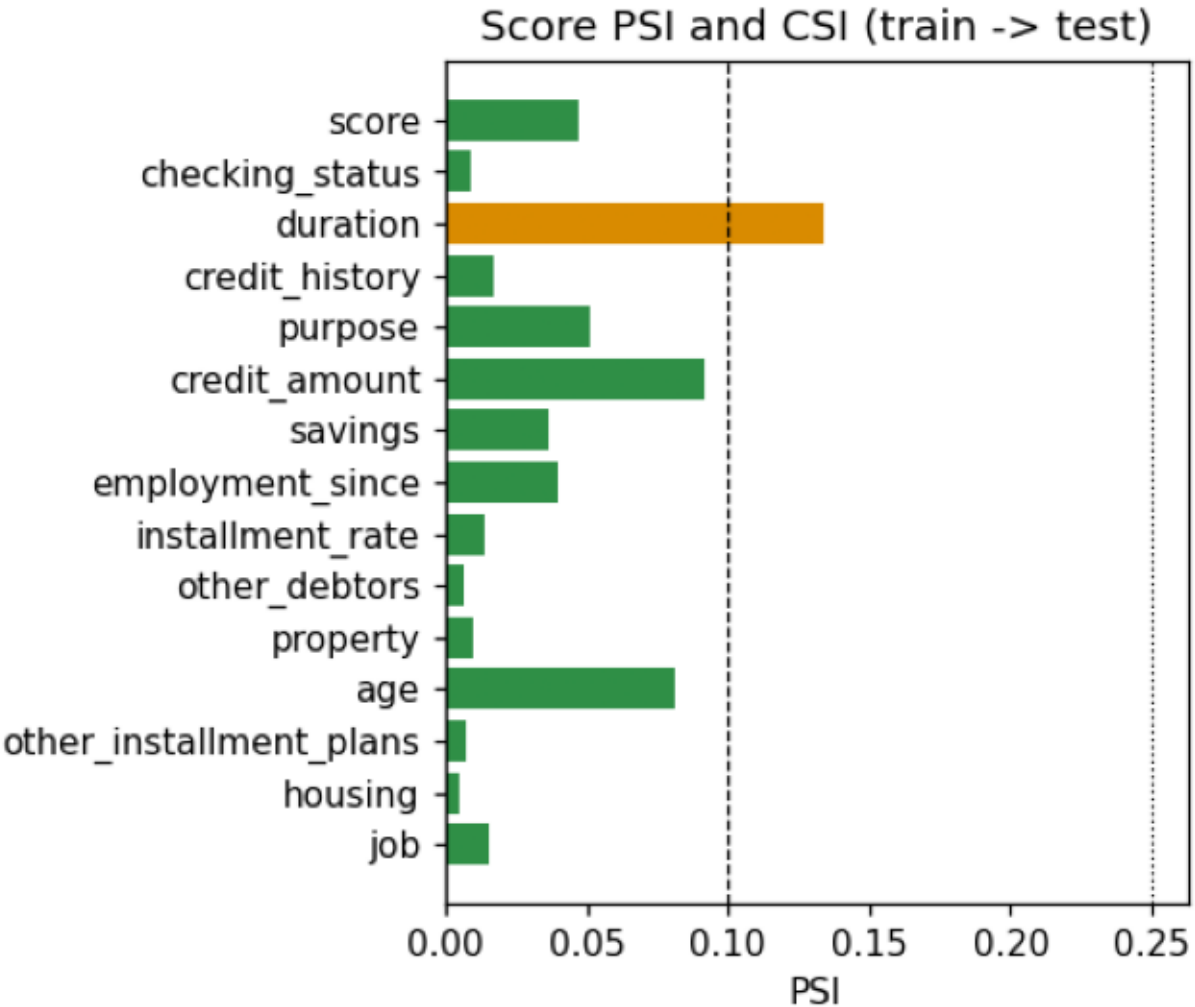
Performance: roc curve



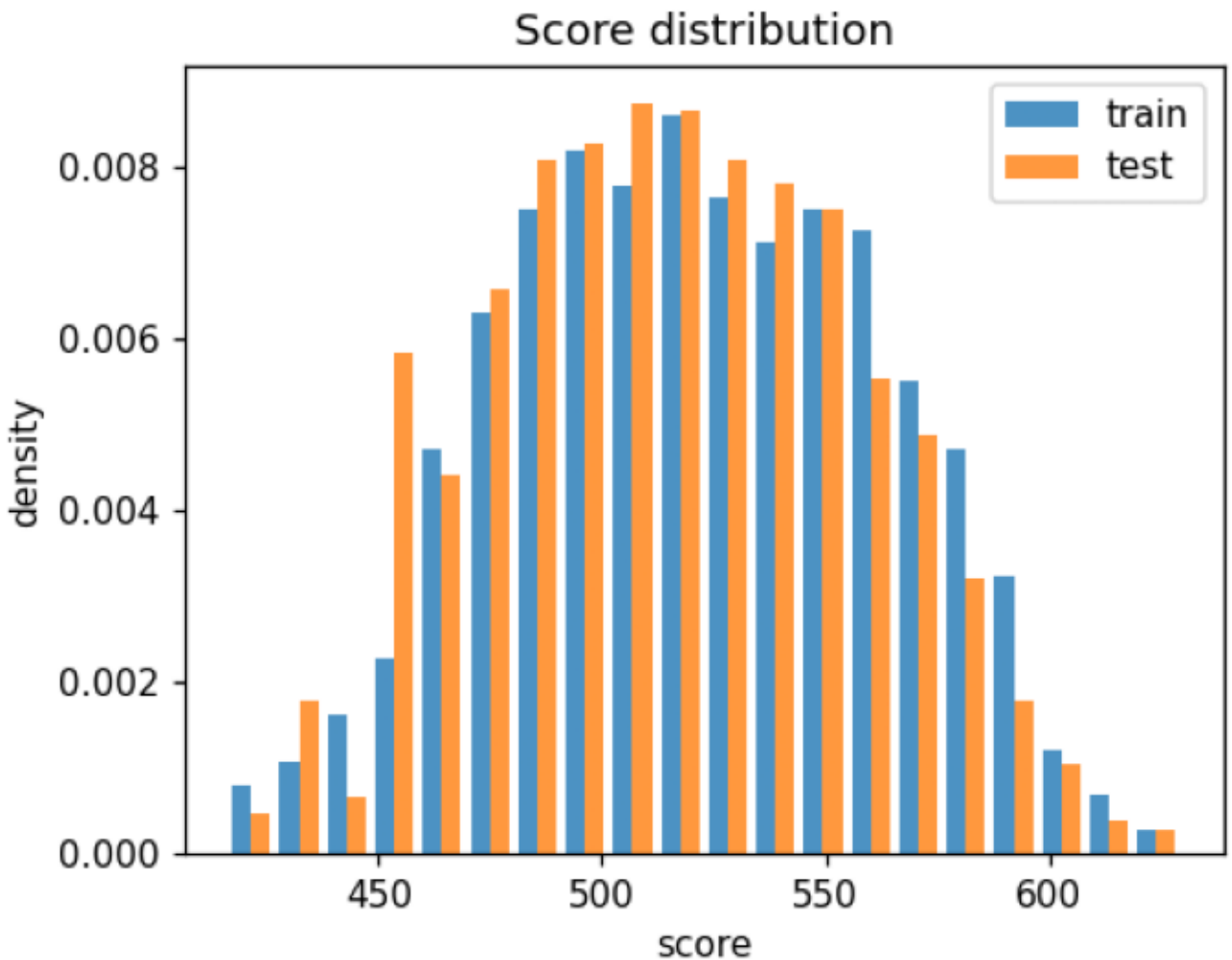
Calibration: calibration



Stability: psi by variable



Stability: score distribution



Fairness: approval rates

